BOND VALUATION

A CASE BOOK

25 Applied Problems in Bond Pricing, Yield, Duration, Convexity and Immunization
Set in the Indian Fixed-Income Market

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A Case Book for MBA / PGDM Programmes in
Financial Management • Fixed Income Securities • Investment Analysis

Preface and Note on Use

This case book contains twenty-five bond-valuation problems, organized into eight thematic parts that follow the natural teaching sequence of a Fixed Income Securities or Investment Analysis course: bond value theorems, term structure and forward rates, bond return decomposition, yield to call, duration, interest-rate risk, immunization, and convexity.

Each case is built around a realistic Indian financial-market setting — a bank treasury, an NBFC, a life insurer, a debt mutual fund, a housing finance company, a primary dealer, or a corporate gratuity trust — while preserving, without alteration, the original academic question, numerical inputs and valuation formula of the source problem it is drawn from. Contextualization is intended to make each problem's financial stakes concrete; it does not change what is being asked, what data are given, or what formula and answer are correct.

Every case follows the same structure: a short institutional context and learning objective, a case situation, one or more required questions, and a six- or seven-part solution — the relevant valuation principle, the formula, a fully worked substitution (with tables wherever they improve transparency), the final result, a financial interpretation, a decision or recommendation where one is warranted, and a closing teaching insight.

All figures are stated in Indian Rupees (₹), following the standard Indian numbering convention (lakhs and crores) where the amounts involved warrant it. Present-value and future-value interest factors follow the standard rounded four-decimal tables customary in Indian finance textbooks; recomputation with an unrounded spreadsheet or financial calculator may therefore differ from the figures shown here by a few paise to a few rupees, without affecting any conclusion. Where a case consolidates or clarifies a point of presentation, this is flagged in an Editorial / Technical Note beneath the relevant case, in keeping with sound academic practice.

The case book is intended for direct classroom use: each problem can be assigned individually, or the eight parts can be taught in sequence as a complete module moving from bond pricing fundamentals through to the advanced interest-rate-risk toolkit of duration, immunization and convexity.

Structure of This Case Book

Part	Topic	Cases
Part I	Bond Value Theorems	*Cases 1–5*
Part II	Term Structure and Forward Rates	*Cases 6–8*
Part III	Bond Return Decomposition	*Cases 9–15*
Part IV	Yield to Call	*Case 16*
Part V	Duration	*Cases 17–19*
Part VI	Interest-Rate Risk	*Cases 20–22*
Part VII	Immunization	*Case 23*
Part VIII	Convexity	*Cases 24–25*

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Part I — Bond Value Theorems

Five cases establishing the foundational relationships between bond price, coupon rate, yield to maturity, maturity, and the direction and size of yield changes.

Case 1: Discount Narrows as Maturity Approaches

Apex Infrastructure Finance Ltd — Corporate Treasury Desk

Learning Objective

To value a bond trading at a discount to face value and to demonstrate that, when the required yield exceeds the coupon rate, the discount progressively narrows as the bond moves toward maturity.

Case Situation

The corporate treasury desk of Apex Infrastructure Finance Ltd, a Mumbai-based NBFC engaged in infrastructure lending, holds a six-year corporate bond of face value ₹5,000 carrying an annual coupon rate of 9%. Interest rates in the corporate bond market have since hardened, and comparable AA-rated paper is now trading to yield 11%.

The desk head has asked the treasury analyst to mark the bond to market at the current required yield of 11%, and to trace how the bond's price would move each year as it runs down to redemption, assuming the market's required yield of 11% stays unchanged. This exercise feeds into the desk's quarterly mark-to-market note for the investment committee.

Required

1. Calculate the bond's price today (with six years remaining to maturity) at a required yield of 11%.
2. Trace how the bond price changes as the remaining maturity falls from 6 years to 0 years, with YTM held constant at 11%.
3. Explain, in terms of the coupon rate and YTM, why the bond trades below face value and why the gap closes over time.

Solution

A. Relevant Valuation Principle

A bond's price is the present value of its promised cash flows — the annuity of coupons plus the lump-sum redemption of face value — discounted at the market's required yield (YTM). When the coupon rate is fixed below the prevailing YTM, investors will only pay less than face value for the bond, since the fixed coupon is less attractive than what comparable new paper offers. This gap between price and face value is the bond's discount.

As the bond approaches maturity, the number of remaining discount periods shrinks, so the present-value drag on the redemption amount lessens each year. The price is pulled steadily toward ₹5,000, reaching exactly face value at maturity — this is the first Bond Value Theorem on the effect of the passage of time.

B. Formula

$$P_0 = C\,(PVIFA\ r,N) + F\,(PVIF\ r,N)$$

where P_0 = current bond price, C = annual coupon, F = face value, r = required yield (YTM), N = years to maturity.

C. Substitution and Working

Given data:

Item	Value
Face value (F)	₹5,000
Annual coupon rate	9%
Annual coupon (C)	₹450
Required yield (YTM)	11%
Remaining maturity (N)	6 years

Substituting into the valuation formula:

$$P_0 = 450\,(PVIFA\ 11\%,6) + 5{,}000\,(PVIF\ 11\%,6)$$

Using standard present-value interest-factor tables, PVIFA(11%,6) ≈ 4.2305 and PVIF(11%,6) ≈ 0.5346, so:

$$P_0 = 450 \times 4.2305 + 5{,}000 \times 0.5346 \approx ₹4{,}576.95$$

Repeating the same calculation for each remaining maturity from 5 years down to 0 years (holding YTM at 11%) gives the following price path:

Remaining Maturity	Bond Price (₹)
6 years	4,576.95
5 years	4,630.40
4 years	4,689.75
3 years	4,755.65
2 years	4,828.75
1 year	4,909.90
0 years (maturity)	5,000.00

D. Final Result

P_0 (6-year discount bond, YTM = 11%) ≈ ₹4,576.95
Discount narrows steadily to ₹0 at maturity

E. Interpretation

The bond is priced below face value because the required yield (11%) exceeds the coupon rate (9%): an investor buying the bond only at ₹5,000 would earn a return equal to the coupon rate of 9%, which is less than what the market currently demands. The market therefore marks the price down until the total return — coupon income plus price appreciation to par — equals 11%.

As the table shows, the discount of ₹423.05 with six years remaining shrinks to ₹369.60, ₹310.25, ₹244.35, ₹171.25, and ₹90.10 in successive years, vanishing entirely at maturity when the bond is redeemed at exactly ₹5,000. This is the well-known 'pull to par' effect.

F. Decision / Recommendation

For Apex Infrastructure Finance's treasury desk, the mark-to-market entry for this holding should reflect a value of approximately ₹4,576.95, not the original issue price of ₹5,000. The desk should also flag that, purely from the passage of time (with YTM unchanged), the bond's book value will appreciate by roughly ₹50–₹80 a year purely from the discount unwinding — a predictable and riskless component of return that is distinct from any view on future interest-rate movements.

G. Teaching Insight

Teaching Insight: When the required yield exceeds the coupon rate, a bond trades at a discount, and that discount shrinks mechanically toward zero as maturity approaches — even if market yields do not change at all. Students should learn to separate this 'pull to par' effect from price changes caused by shifts in the general level of interest rates.

Case 2: Premium Narrows as Maturity Approaches

Apex Infrastructure Finance Ltd — Corporate Treasury Desk

Learning Objective

To value a bond trading at a premium to face value and to demonstrate that, when the coupon rate exceeds the required yield, the premium progressively narrows as the bond moves toward maturity.

Case Situation

Six months later, the interest-rate environment has reversed. The same 9%-coupon, six-year Apex Infrastructure Finance bond is now being evaluated in a falling-rate scenario: comparable corporate paper is yielding only 7%, well below the bond's coupon.

The treasury analyst has been asked to repeat the earlier exercise — value the bond at the new required yield of 7%, and trace the price path as the bond runs down to maturity — so that the investment committee can see how the bond's premium behaves as rates fall and time passes.

Required

1. Calculate the bond's price today (with six years remaining to maturity) at a required yield of 7%.
2. Trace how the bond price changes as remaining maturity falls from 6 years to 0 years, with YTM held constant at 7%.
3. Explain why the bond now trades above face value and why the premium erodes over time.

Solution

A. Relevant Valuation Principle

When a bond's coupon rate exceeds the market's required yield, investors are willing to pay more than face value to capture the above-market coupon stream — the bond trades at a premium. As with a discount bond, the number of remaining periods over which this above-market coupon is received shrinks each year, so the premium is progressively amortised away, and price converges to face value at redemption.

B. Formula

$$P_0 = C\,(PVIFA\ r,N) + F\,(PVIF\ r,N)$$

Same valuation formula as Case 1, now evaluated at YTM = 7%.

C. Substitution and Working

Given data:

Item	Value
Face value (F)	₹5,000
Annual coupon rate	9%
Annual coupon (C)	₹450
Required yield (YTM)	7%
Remaining maturity (N)	6 years

$$P_0 = 450\,(PVIFA\ 7\%,6) + 5{,}000\,(PVIF\ 7\%,6)$$

Using PVIFA(7%,6) ≈ 4.7665 and PVIF(7%,6) ≈ 0.6663:

$$P_0 = 450 \times 4.7665 + 5{,}000 \times 0.6663 \approx ₹5{,}476.65$$

The price path as maturity approaches (YTM held at 7%) is:

Remaining Maturity	Bond Price (₹)
6 years	5,476.65
5 years	5,410.00
4 years	5,338.70
3 years	5,262.45
2 years	5,180.80
1 year	5,093.45
0 years (maturity)	5,000.00

D. Final Result

P_0 (6-year premium bond, YTM = 7%) ≈ ₹5,476.65
Premium narrows steadily to ₹0 at maturity

E. Interpretation

The bond now trades at a premium of ₹476.65 because its 9% coupon exceeds the 7% yield available on comparable paper. Investors bid the price up so that the total return — the now-below-market coupon relative to price, plus the capital loss as price falls back to ₹5,000 at maturity — works out to exactly 7%.

The premium erodes from ₹476.65 to ₹410.00, ₹338.70, ₹262.45, ₹180.80, and ₹93.45 in successive years before vanishing at redemption. Unlike the discount case, this 'pull to par' now works against the holder: even with yields unchanged, the bond's price is expected to fall each year purely from the passage of time.

F. Decision / Recommendation

Apex Infrastructure Finance's treasury desk should recognise that holding this bond to maturity in the current environment will generate a capital loss of ₹476.65 that is already priced in and predictable — it is not a sign of credit deterioration. If the desk needed to raise funds, selling now versus later should be evaluated net of this scheduled price decline, not against the original purchase price alone.

G. Teaching Insight

Teaching Insight: When the coupon rate exceeds the required yield, a bond trades at a premium that shrinks mechanically to zero as maturity approaches. Read together, Cases 1 and 2 show that regardless of whether a bond starts at a discount or a premium, its price always converges to face value at maturity — the first and most fundamental Bond Value Theorem.

Case 3: Maturity and Price Sensitivity to Yield Changes

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To demonstrate that, for a given change in YTM, longer-maturity bonds experience a greater percentage change in price than shorter-maturity bonds — the foundation of interest-rate (duration) risk.

Case Situation

The investments desk at Konkan Bank Ltd holds two AAA-rated corporate bonds in its held-for-trading book, both carrying an 8% coupon and currently priced at par (YTM = 8%): Bond A, maturing in 4 years, and Bond B, maturing in 9 years.

Ahead of the RBI's next monetary policy review, the bank's ALM committee wants to understand how much more the longer bond would fall in value than the shorter one if market yields moved up by just 1 percentage point, to gauge which position carries greater interest-rate risk.

Required

1. Calculate the price of Bond A and Bond B at the current YTM of 8% and at a revised YTM of 9%.

2. Calculate the percentage change in price for each bond.
3. Explain why the longer-maturity bond is more sensitive to the yield change.

Solution

A. Relevant Valuation Principle

Bond price is the sum of discounted cash flows, and each cash flow's present value is more sensitive to the discount rate the further away in time it lies (present value falls geometrically, not linearly, with time). A longer-maturity bond therefore has proportionately more of its value tied up in distant, rate-sensitive cash flows, making its price more elastic with respect to yield changes than a shorter bond with the same coupon and starting yield.

B. Formula

$$P_0 = C\ (PVIFA\ r,N) + F\ (PVIF\ r,N)$$

$$\%\ Change\ in\ Price = [(P\ at\ old\ YTM - P\ at\ new\ YTM)\ /\ P\ at\ old\ YTM] \times 100$$

C. Substitution and Working

Given data:

Particulars	Bond A	Bond B
Face value	₹1,000	₹1,000
Coupon rate	8%	8%
Maturity	4 years	9 years
Initial YTM	8%	8%
Revised YTM	9%	9%

At YTM = 8%, coupon rate equals YTM for both bonds, so both are priced exactly at face value:

$$P_0\ (Bond\ A) = P_0\ (Bond\ B) = ₹1{,}000$$

At YTM = 9%, revaluing each bond over its own remaining maturity gives:

	Bond A (4 yrs)	Bond B (9 yrs)
Price at YTM = 8%	₹1,000.00	₹1,000.00
Price at YTM = 9%	₹967.60	₹940.05
% change in price	3.24%	5.99%

Bond A: % Change = (1,000 − 967.60) / 1,000 × 100 = 3.24%

Bond B: % Change = (1,000 − 940.05) / 1,000 × 100 = 5.99%

D. Final Result

Bond A (4-year): price falls 3.24% for a 1-point rise in YTM
Bond B (9-year): price falls 5.99% for the same 1-point rise in YTM

E. Interpretation

Both bonds share the same coupon rate, face value, and the same 1-percentage-point rise in required yield — the only difference is maturity. Bond B, with more than double the remaining life of Bond A, absorbs nearly double the percentage price decline (5.99% versus 3.24%).

This confirms the Bond Value Theorem: for a given change in YTM, the longer the term to maturity, the greater the change in price. Maturity (and, more precisely, duration) is therefore a direct measure of a bond's exposure to interest-rate movements.

F. Decision / Recommendation

If Konkan Bank's ALM committee expects the RBI to tighten policy, the treasury desk should recognise that Bond B contributes materially more mark-to-market risk to the trading book per rupee invested than Bond A. Position limits, or a partial switch from the 9-year into the 4-year bond, would reduce the book's sensitivity to an upward yield shock, at the cost of forgoing some carry from the longer paper.

G. Teaching Insight

Teaching Insight: Holding coupon rate and the size of the yield change constant, longer maturity always means greater price volatility. This is the intuitive precursor to duration, which formalises maturity-driven interest-rate risk with more precision.

Case 4: The Asymmetry of Bond Price Changes

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To demonstrate that, for a given bond, an equal-sized fall in YTM produces a larger percentage price gain than the percentage price loss caused by an equal-sized rise in YTM — the practical face of bond price convexity.

Case Situation

As part of the same ALM stress-testing exercise, Konkan Bank's risk team wants to test a single seven-year, 8%-coupon bond (currently priced at par, since YTM also stands at 8%) under two symmetric yield shocks: a 2-percentage-point rise to 10%, and a 2-percentage-point fall to 6%.

The risk team wants to know whether the bond would gain and lose the same percentage of value under these opposite but equal-sized shocks, since this has a direct bearing on how the bank models tail risk in its interest-rate scenarios.

Required

1. Calculate the bond price and percentage price decrease when YTM rises from 8% to 10%.
2. Calculate the bond price and percentage price increase when YTM falls from 8% to 6%.

3. Compare the magnitude of the two percentage changes and explain the difference.

Solution

A. Relevant Valuation Principle

Bond price is a convex, not a linear, function of yield: the price-yield relationship curves upward as yields fall. Because of this convexity, the price gain from a given fall in yield is always larger than the price loss from an equal-sized rise in yield, even though the change in YTM is symmetric in both directions.

B. Formula

$$P_0 = C\,(PVIFA\ r,N) + F\,(PVIF\ r,N)$$

C. Substitution and Working

Given data: Face value = ₹1,000; coupon rate = 8% (annual coupon ₹80); maturity = 7 years; current YTM = 8%; current price = ₹1,000.

Scenario 1 — YTM rises from 8% to 10%:

$$P_{10}\% = 80\,(PVIFA\ 10\%,7) + 1{,}000\,(PVIF\ 10\%,7) \approx ₹902.63$$

$$\%\ Decrease = (1{,}000 - 902.63) / 1{,}000 \times 100 = 9.74\%$$

Scenario 2 — YTM falls from 8% to 6%:

$$P_6\% = 80\,(PVIFA\ 6\%,7) + 1{,}000\,(PVIF\ 6\%,7) \approx ₹1{,}111.65$$

$$\%\ Increase = (1{,}111.65 - 1{,}000) / 1{,}000 \times 100 = 11.17\%$$

Change in YTM	Bond Price (₹)	% Change
8% → 10%	902.63	−9.74%
8% → 6%	1,111.65	+11.17%

D. Final Result

A 2-point rate rise costs 9.74% of value
An equal 2-point rate fall adds 11.17% of value
11.17% > 9.74%: the gain exceeds the loss

E. Interpretation

A 2-percentage-point decrease in YTM produces a larger percentage price gain (11.17%) than the percentage loss (9.74%) caused by an equal 2-percentage-point increase in YTM. The bond's price does not respond symmetrically to equal but opposite yield shocks.

This asymmetry is the direct consequence of the bond value theorem that bond prices are convexly, not linearly, related to yield — a property formally measured later in this casebook as convexity (see Cases 24-25).

F. Decision / Recommendation

For Konkan Bank's risk team, this means duration-based linear approximations of price risk will systematically overstate losses in a rate sell-off and understate gains in a rate rally. Any VaR or stress-loss model built purely on modified duration should be treated as a conservative (upper-bound) estimate of downside risk, and refined with a convexity adjustment for large yield moves.

G. Teaching Insight

Teaching Insight: Given the maturity, the price change from a decrease in YTM is always larger than the price change from an equal increase in YTM. This asymmetry — bond price convexity — is a structural feature of fixed-income securities, not a data quirk, and it favours the long bondholder in volatile markets.

Case 5: Coupon Rate and Price Sensitivity

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To demonstrate that, for a given maturity and a given change in YTM, a higher-coupon bond experiences a smaller percentage price change than a lower-coupon bond.

Case Situation

A fund manager at Himalaya Debt Advantage Fund, an open-ended Indian debt mutual fund, is comparing two five-year corporate bonds of identical face value and maturity for the fund's accrual portfolio: Bond A carries an 8% coupon, and Bond B carries an 11% coupon.

Both bonds currently yield to their respective coupon rates. The fund manager wants to know which bond would lose more value in percentage terms if the market's required yield rose uniformly to 10% for both — information that matters for choosing between the two for a portfolio that may need to be rebalanced before maturity.

Required

1. Calculate the price of Bond A and Bond B at YTM = 8% and at YTM = 10%.
2. Calculate the percentage change in price for each bond.
3. Explain why the higher-coupon bond is less sensitive to the change in yield.

Solution

A. Relevant Valuation Principle

A higher coupon rate returns a greater share of the bond's total value earlier, through larger periodic coupons, so proportionately less of the bond's value depends on the single, distant, and most rate-sensitive redemption payment. All else equal, a higher-coupon bond therefore has a shorter effective (duration-weighted) life than a lower-coupon bond of the same maturity, and so its price responds less to a given change in yield.

B. Formula

$$P_0 = C\,(PVIFA\ r,N) + F\,(PVIF\ r,N)$$

$$\% \text{ Change in Price} = [(P \text{ at old YTM} - P \text{ at new YTM}) / P \text{ at old YTM}] \times 100$$

C. Substitution and Working

Given data:

Particulars	Bond A	Bond B
Face value	₹1,000	₹1,000
Coupon rate	8%	11%
Maturity	5 years	5 years
Initial YTM	8%	8%
New YTM	10%	10%

Step 1 — Price at YTM = 8%. Bond A's coupon equals YTM, so it prices at par; Bond B's coupon exceeds YTM, so it prices at a premium:

$$P\_A = ₹1{,}000.00 \qquad P\_B = ₹1{,}119.78$$

Step 2 — Price at YTM = 10%:

$$P\_A = ₹924.18 \qquad P\_B = ₹1{,}037.91$$

Step 3 — Percentage change in price:

$$\text{Bond A: } \% \text{ Change} = (1{,}000 - 924.18) / 1{,}000 \times 100 = 7.58\%$$

$$\text{Bond B: } \% \text{ Change} = (1{,}119.78 - 1{,}037.91) / 1{,}119.78 \times 100 \approx 7.31\%$$

	Bond A	Bond B
Coupon rate	8%	11%
Price at YTM = 8%	₹1,000.00	₹1,119.78
Price at YTM = 10%	₹924.18	₹1,037.91
% change in price	7.58%	7.31%

D. Final Result

Lower-coupon Bond A: price falls 7.58%
Higher-coupon Bond B: price falls only 7.31% for the same yield shock

E. Interpretation

Bond B, with the higher 11% coupon, delivers more of its cash flow sooner. This makes it marginally less sensitive to the identical 2-percentage-point rise in YTM than Bond A, even though both bonds share the same face value and five-year maturity.

The difference (7.58% versus 7.31%) is modest here because the maturity is short, but the effect widens considerably for longer-dated bonds, where the redemption payment is a much larger share of present value.

F. Decision / Recommendation

For Himalaya Debt Advantage Fund, if the manager expects yields to rise and wants to limit mark-to-market drawdown in the accrual book without shortening maturity, the higher-coupon Bond B is the marginally more defensive holding. Conversely, in a falling-rate view, the lower-coupon Bond A would show slightly larger price appreciation for the same maturity.

G. Teaching Insight

Teaching Insight: For a given change in YTM, the percentage price change of a high-coupon bond is smaller than that of a low-coupon bond, other things being equal. Coupon rate is therefore an independent lever — alongside maturity — that portfolio managers use to fine-tune a bond's interest-rate sensitivity.

Part II — Term Structure and Forward Rates

Three cases on extracting forward rates from the spot-rate curve, and on the classic yield-curve-riding strategy for money-market instruments.

Case 6: Extracting Forward Rates from the G-Sec Spot Curve

Primary Dealer Desk, Government Securities Trading Desk

Learning Objective

To derive the market's implied one-year forward rates from a set of observed spot yields on Central Government Securities, using the no-arbitrage relationship between spot and forward rates.

Case Situation

The government securities trading desk of a primary dealer is building its term-structure model for the coming quarter. The desk observes the following spot yields on Central Government Securities: the 1-year security yields 7.20%, the 2-year security yields 7.80%, and the 3-year security yields 8.60%.

Before quoting forward-starting swaps and advising institutional clients on curve positioning, the desk's junior dealer has been asked to back out the one-year forward rates implied by this spot curve — the rates the market is effectively pricing for future one-year borrowing and lending.

Required

1. Calculate the one-year forward rate beginning one year from now, f_2.
2. Calculate the one-year forward rate beginning two years from now, f_3.

Solution

A. Relevant Valuation Principle

Under the pure expectations / no-arbitrage view of the term structure, investing for n years at the n-year spot rate must yield the same terminal wealth as rolling over a sequence of one-year investments at today's one-year rate and the market's implied forward rates for each subsequent year. Rearranging this identity isolates each forward rate.

B. Formula

$$(1 + r_n)^n = (1 + r_1)(1 + f_2) \cdots (1 + f_n)$$

C. Substitution and Working

Given data:

Maturity	Spot Rate
1 year (r_1)	7.20%
2 years (r_2)	7.80%

Maturity	Spot Rate
3 years (r_3)	8.60%

Step 1 — Solve for f_2 using the two-year relationship:

$$(1 + r_2)^2 = (1 + r_1)(1 + f_2) \implies f_2 = (1 + r_2)^2 / (1 + r_1) - 1$$

$$f_2 = (1.078)^2 / (1.072) - 1 \approx 8.40\%$$

Step 2 — Solve for f_3 using the three-year relationship, now feeding in the f_2 just derived:

$$(1 + r_3)^3 = (1 + r_1)(1 + f_2)(1 + f_3) \implies f_3 = (1 + r_3)^3 / [(1 + r_1)(1 + f_2)] - 1$$

$$f_3 = (1.086)^3 / [(1.072)(1.0840)] - 1 \approx 10.22\%$$

Forward Rate	Value
f_2 (year 1 → year 2)	8.40%
f_3 (year 2 → year 3)	10.22%

D. Final Result

$f_2 \approx 8.40\%$
$f_3 \approx 10.22\%$

E. Interpretation

The forward rate is the future one-year interest rate that is consistent with — and implied by — today's observed spot-rate structure; it is not a forecast published by any authority but a rate backed out algebraically so that no arbitrage opportunity exists between investing long and rolling over short.

Here, f_2 (8.40%) and especially f_3 (10.22%) lie well above the current 1-year and 2-year spot rates, signalling that the market is pricing in materially higher short-term rates in the future — consistent with the sharply upward-sloping, steepening spot curve (7.20% → 7.80% → 8.60%).

F. Decision / Recommendation

The desk can use f_2 and f_3 as reference rates for pricing forward-starting government-security repos or interest-rate swaps with institutional clients, and as a cross-check against the desk's independent view on the RBI's future rate path. A forward rate well above the current spot curve is often read as the market pricing in future rate hikes or an inflation risk premium.

G. Teaching Insight

Teaching Insight: A sequence of spot rates can always be decomposed into a sequence of implied forward rates using the compounding identity $(1+r_n)^n = (1+r_1)(1+f_2)\cdots(1+f_n)$. This is the basic building block for pricing forward-starting instruments and for reading the market's implied expectations of future short-term rates.

Case 7: The Implied Forward Rate for Year 5

Setu Housing Finance Corp Ltd — Corporate Treasury

Learning Objective

To derive a single implied forward rate from just two spot rates, using the shortcut ratio method — a common requirement when a firm is deciding on the timing of future borrowing.

Case Situation

The corporate treasury of Setu Housing Finance Corp Ltd is evaluating the cost of a bullet borrowing it plans to raise four years from now, to be repaid one year later (i.e., a one-year loan starting in year 5). The treasury team observes that five-year bonds in the market currently yield 7.6%, while four-year bonds yield 6.8%.

Before approaching lenders, the treasurer wants to estimate the one-year borrowing rate the market is implicitly pricing for that future window, so the company can judge whether locking in a forward-rate agreement today would be cheaper than waiting and borrowing at the prevailing spot rate in four years.

Required

1. Calculate the implied one-year forward rate, f_5, for a one-year investment/borrowing beginning four years from now.

Solution

A. Relevant Valuation Principle

When only the n-year and (n−1)-year spot rates are available, the forward rate for the final (nth) year can be extracted directly, without needing every intermediate forward rate, by dividing the compounded n-year return by the compounded (n−1)-year return.

B. Formula

$$(1 + r_5)^5 = (1 + r_1)(1 + f_2)(1 + f_3)(1 + f_4)(1 + f_5)$$

$$(1 + r_4)^4 = (1 + r_1)(1 + f_2)(1 + f_3)(1 + f_4)$$

$$\Longrightarrow f_5 = (1 + r_5)^5 / (1 + r_4)^4 - 1$$

C. Substitution and Working

Given data: four-year spot rate r_4 = 6.8%; five-year spot rate r_5 = 7.6%; required forward period = Year 5.

Dividing the five-year compounding identity by the four-year identity cancels out r_1, f_2, f_3 and f_4, leaving:

$$f_5 = (1 + r_5)^5 / (1 + r_4)^4 - 1$$

$$f_5 = (1.076)^5 / (1.068)^4 - 1 \approx 10.86\%$$

D. Final Result

$f_5 \approx 10.86\%$

E. Interpretation

The market is implicitly pricing a one-year rate of about 10.86% for borrowing that begins four years from now — considerably higher than either the current 4-year (6.8%) or 5-year (7.6%) spot rate. This again reflects an upward-sloping and steepening curve: each additional year of maturity is compensated by a progressively higher implied forward rate.

F. Decision / Recommendation

Setu Housing Finance's treasury should compare this implied forward rate of 10.86% against its own internal view of where one-year rates are likely to stand four years from now. If management believes actual rates in year 5 are likely to be lower than 10.86%, it may prefer to wait and borrow at the spot rate then, rather than locking in a forward-rate agreement or issuing a longer-dated bond today at rates that already embed this higher forward cost.

G. Teaching Insight

Teaching Insight: $f_n = (1+r_n)^n / (1+r_{n-1})^{n-1} - 1$ is a convenient shortcut whenever the required forward period is the final year of the sequence — it avoids computing every intermediate forward rate and is especially useful for a treasury team assessing a single future funding decision.

Case 8: Riding an Upward-Sloping Yield Curve with Treasury Bills

Himalaya Debt Advantage Fund — Money Market Desk

Learning Objective

To evaluate the 'riding the yield curve' strategy — buying a longer-dated money-market instrument and selling it before maturity to capture price appreciation from rolling down an upward-sloping curve.

Case Situation

The money-market desk of Himalaya Debt Advantage Fund is deciding how to deploy short-term surplus cash. Two T-bills are quoted: a 91-day T-bill priced at ₹97.80 and a 182-day T-bill priced at ₹95.20 (both per ₹100 face value).

Rather than simply buying and holding the 91-day bill to its own maturity, the desk is considering the classic 'riding the yield curve' trade: buy the 182-day bill today, and sell it after 91 days — at which point it will have exactly 91 days left to run, and should, if the yield curve is unchanged, be priced like a fresh 91-day bill, i.e. at ₹97.80.

Required

1. Calculate the investor's annualised holding-period yield from riding the curve (buying the 182-day bill and selling it after 91 days at an assumed price of ₹97.80).
2. Explain the strategy and the condition under which it works.

Solution

A. Relevant Valuation Principle

'Riding the yield curve' is a buy-and-hold-then-sell strategy: an investor buys a bond or bill further out on an upward-sloping curve (where yields, and hence discount, are higher) and sells it before maturity, once its remaining life has shortened to a point on the curve with a lower yield. If the curve's shape is stable, the bill's price rises purely from 'sliding down' to a lower-yield point, giving the investor a return that exceeds the yield originally quoted on the shorter bill.

B. Formula

$$\text{Holding-Period Return (HPR)} = (\text{Selling Price} - \text{Purchase Price}) / \text{Purchase Price}$$

$$\text{Annualised Yield} = \text{HPR} \times (365 / \text{Holding Period in Days})$$

C. Substitution and Working

Given data:

Particular	Value
Purchase price (182-day bill today)	₹95.20
Expected selling price (after 91 days)	₹97.80
Holding period	91 days

$$HPR = (97.80 - 95.20) / 95.20 = 0.027311, \text{ i.e. } 2.73\%$$

Annualising over a 365-day year:

$$\text{Annualised Yield} = (97.80 - 95.20) / 95.20 \times (365 / 91) \approx 10.95\%$$

D. Final Result

Annualised holding-period yield from riding the curve ≈ 10.95%

E. Interpretation

For reference, the quoted yield on the 91-day bill itself is about 9.02%, while the 182-day bill's own yield to its own maturity is about 10.11%. By buying the longer, higher-yielding 182-day bill and selling it after 91 days — once it has 'become' a 91-day bill — the investor captures an annualised return of 10.95%, higher than either bill's own quoted yield.

This extra return comes purely from the roll-down of the yield curve: as the bill's remaining life shortens from 182 to 91 days, its required yield falls (since the curve is upward-sloping), and its price rises accordingly, over and above the pure passage-of-time pull to par.

F. Decision / Recommendation

The strategy is attractive for Himalaya Debt Advantage Fund's money-market desk only so long as the yield curve retains its current upward slope and shape over the 91-day holding period. Should short-term rates rise unexpectedly, or the curve flatten or invert, the actual selling price after 91 days could fall short of the assumed ₹97.80, eroding or reversing the strategy's advantage over simply holding the 91-day bill to maturity. The desk should therefore treat riding the curve as an active curve-shape bet, not a riskless yield pickup.

G. Teaching Insight

Teaching Insight: Riding the yield curve enhances return only when the curve's upward slope persists over the holding period; the strategy substitutes reinvestment-rate risk (from holding short paper) for curve-shape risk (that the curve flattens before the position is sold), and students should be able to identify this trade-off explicitly.

Part III — Bond Return Decomposition

Seven linked cases building up an investor's total bond return — coupon income, interest-on-interest, and capital gain — and converting it into a realized yield across different holding periods and reinvestment-rate scenarios.

Case 9: Interest-on-Interest and the Reinvestment Rate

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute the interest-on-interest (IoI) component of a bond's total return, and to show how it shrinks when the reinvestment rate available on coupon receipts falls below the bond's own yield to maturity.

Case Situation

Suraksha Life Insurance Co. Ltd, a mid-sized Indian life insurer, holds a 12% annual-coupon corporate bond of face value ₹500 in its policyholders' fund. The bond is redeemable after six years at a 4% premium (i.e., at ₹520), and was purchased at ₹470.23, giving it a yield to maturity of approximately 14%.

The insurer's investment team reinvests every coupon it receives in similar short-tenor instruments. Initially, coupons can be reinvested at 14%, matching the bond's own YTM. The team now wants to test a scenario in which the interest-rate environment eases and the reinvestment rate available on incoming coupons falls to 11%, to understand how much of the bond's total return this reinvestment risk could take away by the time the bond matures in six years.

Required

1. Calculate the interest-on-interest (IoI) earned over the bond's six-year life if coupons are reinvested at 14%.
2. Calculate the IoI if coupons are instead reinvested at 11%.
3. Compare the two results and comment on the effect of a falling reinvestment rate.

Solution

A. Relevant Valuation Principle

A bond's total return is not limited to the coupons themselves: each coupon received before maturity can be reinvested, and the interest earned on those reinvested coupons — over and above the coupons' own face value — is called interest-on-interest (IoI). Because IoI depends on a rate the investor does not control at the time of purchase (the future reinvestment rate), it is a genuine source of return uncertainty, distinct from the bond's quoted YTM.

B. Formula

$$IoI = C\,(FVIFA\ r,N) - C{\cdot}N$$

$$\text{equivalently: } IoI = C\,[(FVIFA\ r,N) - N]$$

where C = annual coupon, r = reinvestment rate, N = number of years coupons are received and reinvested, and FVIFA is the future-value interest factor for an annuity.

C. Substitution and Working

Given data: Face value = ₹500; coupon rate = 12%, so annual coupon C = ₹60; maturity = 6 years; redemption premium = 4%; purchase price = ₹470.23; YTM ≈ 14%.

Scenario 1 — Reinvestment rate = 14% (equal to the bond's own YTM):

$$IoI = 60\ (FVIFA\ 14\%,6) - (6 \times 60) = 512.13 - 360.00 = ₹152.13$$

Scenario 2 — Reinvestment rate = 11%:

$$IoI = 60\ (FVIFA\ 11\%,6) - (6 \times 60) = 474.77 - 360.00 = ₹114.77$$

Reinvestment Rate	Interest-on-Interest
14%	₹152.13
11%	₹114.77
Reduction	**₹37.36**

D. Final Result

IoI at 14% reinvestment ≈ ₹152.13
IoI at 11% reinvestment ≈ ₹114.77
Fall in reinvestment rate costs ₹37.36 of return

E. Interpretation

A decline in the reinvestment rate from 14% to 11% reduces the interest-on-interest component by ₹37.36 over the bond's six-year life — a meaningful erosion given that the bond's total coupon income over the period is only ₹360 (6 years × ₹60).

This demonstrates that even a bond with a fixed, contractually guaranteed coupon does not guarantee a fixed total return: the realised outcome depends partly on the rate at which interim coupon receipts can be redeployed, a risk known as reinvestment risk.

F. Decision / Recommendation

For Suraksha Life Insurance, this quantifies exactly how much of the bond's promised return is at risk if market rates fall over the holding period — information the actuarial and investment teams need when matching this asset against fixed policyholder liabilities. A falling-rate scenario helps the insurer, since the bond's own price rises, but it simultaneously hurts the coupon-reinvestment leg of return (see Cases 11-13 for how these two effects combine).

G. Teaching Insight

Teaching Insight: Reinvestment Rate ↑ ⟹ Interest-on-Interest ↑, and Reinvestment Rate ↓ ⟹ Interest-on-Interest ↓. IoI is the channel through which reinvestment risk enters a bond's realised return, and it can only be assessed once a reinvestment-rate assumption is specified — it is never fixed by the bond's coupon alone.

Case 10: Interest-on-Interest Across Different Holding Periods

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To show how the interest-on-interest component builds up as the holding period lengthens, and how this build-up differs at two different reinvestment rates.

Case Situation

Continuing with the same bond, Suraksha Life Insurance's investment team wants a fuller picture: rather than looking only at the full six-year IoI figure, they want to see how the IoI accumulates year by year, for both the 14% and 11% reinvestment scenarios, so the insurer can assess exposure to reinvestment risk at each potential exit horizon.

Required

1. Determine the interest-on-interest component for holding periods of 1, 2, 3, 4, 5 and 6 years, at reinvestment rates of 14% and 11%.
2. Comment on how IoI behaves as the holding period lengthens.

Solution

A. Relevant Valuation Principle

IoI accumulates because each coupon, once received, is reinvested for the remaining portion of the holding period. The first coupon (received in year 1) is reinvested only if the bond is held beyond year 1; the longer the total holding period, the more coupons there are, and the longer each one compounds — so IoI grows with the holding period, and grows faster the higher the reinvestment rate.

B. Formula

$$IoI(n) = C\,(FVIFA\ r,n) - C\cdot n$$

C. Substitution and Working

Applying the same formula as Case 9 for each holding period n = 1 through 6, at reinvestment rates of 14% and 11%:

Holding Period	IoI at 14%	IoI at 11%
1 year	₹0.00	₹0.00
2 years	₹8.40	₹6.60
3 years	₹26.38	₹20.53

Holding Period	IoI at 14%	IoI at 11%
4 years	₹55.27	₹42.58
5 years	₹96.61	₹73.67
6 years	₹152.13	₹114.77

D. Final Result

IoI rises with holding period at both reinvestment rates
IoI at 14% exceeds IoI at 11% at every horizon beyond year 1

E. Interpretation

Two relationships stand out. First, holding period and IoI move together: with no coupon reinvested for even a full year until year 2, IoI is zero at a one-year holding period and grows steadily thereafter, since coupons have progressively more time to compound. Second, at every holding period beyond one year, IoI at the 14% reinvestment rate exceeds IoI at 11%, and the gap widens with time (from ₹1.80 at year 2 to ₹37.36 at year 6).

The practical implication is that reinvestment risk is not a fixed-percentage haircut on return — it compounds, and it matters progressively more the longer an investor intends to hold the bond.

F. Decision / Recommendation

Suraksha Life Insurance should recognise that if it intends to hold this bond close to its full six-year term, the sensitivity of total return to the prevailing reinvestment rate is far larger than if it plans a short holding period — a one-year holding period carries no reinvestment risk on this bond at all, since no coupon has yet been reinvested.

G. Teaching Insight

Teaching Insight: Holding Period ↑ ⟹ IoI ↑, and the sensitivity of IoI to the reinvestment rate also grows with the holding period. This is why reinvestment risk is generally regarded as more material for buy-and-hold, long-horizon investors than for short-horizon traders.

Case 11: Capital Gain from Selling a Bond Before Maturity

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute the market price and resulting capital gain if the bond is sold before maturity, and to show how this capital gain builds up as the holding period lengthens, assuming the required return stays at 14%.

Case Situation

Suraksha Life Insurance Co. Ltd, a mid-sized Indian life insurer, holds a 12% annual-coupon corporate bond of face value ₹500 in its policyholders' fund. The bond is redeemable after six years at a 4% premium (i.e., at ₹520), and was purchased at ₹470.23, giving it a yield to maturity of approximately 14%.

If held to maturity, the bond's redemption value of ₹520 is already known with certainty. But insurers frequently rebalance portfolios before maturity, and the investment team wants to know what the bond would fetch — and what capital gain it would represent over the ₹470.23 purchase price — if sold at various points before the six-year maturity, assuming the required return in the market stays at 14%.

Required

1. Calculate the market price of the bond after one year, assuming the required return remains 14%.
2. Calculate the resulting capital gain over the ₹470.23 purchase price.
3. Extend the calculation to holding periods of 2, 3, 4, 5 and 6 years, and comment on the pattern.

Solution

A. Relevant Valuation Principle

When a bond is sold before maturity, the price realised is the present value, at the prevailing required return, of the bond's remaining cash flows — not the eventual redemption value. As the remaining term shortens, this market price converges toward the redemption value, generating a capital gain (or loss) relative to the original purchase price, over and above the coupon income already received.

B. Formula

$$P_t = C\ (PVIFA\ r, N{-}t) + Redemption\ Value\ (PVIF\ r, N{-}t)$$

$$Capital\ Gain = P_t - P_0$$

C. Substitution and Working

Given: Face value = ₹500; annual coupon C = ₹60; redemption value = 500 × 1.04 = ₹520; purchase price P_0 = ₹470.23; required return = 14%; original maturity = 6 years.

Step 1 — Market price after one year (five years remaining to maturity):

$$P_1 = 60\ (PVIFA\ 14\%,5) + 520\ (PVIF\ 14\%,5) \approx ₹476.06$$

Step 2 — Capital gain after one year:

$$Capital\ Gain = 476.06 - 470.23 = ₹5.83$$

Repeating the same present-value calculation for each holding period from 1 to 6 years (required return held at 14%):

Holding Period	Purchase Price	Market Price	Capital Gain
1 year	₹470.23	₹476.06	₹5.83
2 years	₹470.23	₹482.70	₹12.48
3 years	₹470.23	₹490.28	₹20.06
4 years	₹470.23	₹498.92	₹28.70
5 years	₹470.23	₹508.77	₹38.55
6 years	₹470.23	₹520.00	₹49.77

D. Final Result

Capital gain after 1 year (required return 14%) = ₹5.83
Capital gain grows to ₹49.77 by maturity (year 6)

E. Interpretation

Because the market price is discounted at 14% — above the bond's 12% coupon — the bond trades at a discount to its eventual redemption value of ₹520 throughout its life, but that discount narrows every year as the remaining term shortens, exactly as in the bond value theorems of Cases 1-2. The capital gain therefore grows from ₹5.83 after one year to the full ₹49.77 by maturity, when the market price and redemption value converge exactly.

F. Teaching Insight

Teaching Insight: When the required return exceeds a bond's coupon rate, selling before maturity still realises a capital gain — just a smaller one than holding to redemption — because the bond's price is on a rising trajectory toward the redemption value throughout its life.

Case 12: Capital Gain When the Required Return Falls

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To show how a fall in the market's required return magnifies the capital gain available from selling a bond before maturity, illustrating the inverse relationship between yields and bond prices.

Case Situation

Using the same bond, the investment team now asks: what if interest rates fall and the required return applicable to this bond drops from 14% to 11%? The team wants the market price and capital gain recalculated for holding periods of one to six years under this lower required return, to compare against the 14% scenario already computed in Case 11.

Required

1. Calculate the market price and capital gain for holding periods of 1 to 6 years, assuming the required return is 11%.
2. Compare the results with the 14% scenario and explain the difference.

Solution

A. Relevant Valuation Principle

Bond price and required return move inversely. A fall in the required return raises the present value of all future cash flows, including the eventual redemption payment — so the same bond, sold at the same point in time, will fetch a materially higher price (and larger capital gain) when the required return is lower.

B. Formula

$$P_t = C\,(PVIFA\ r, N-t) + Redemption\ Value\ (PVIF\ r, N-t)$$

C. Substitution and Working

Repeating the Case 11 calculation with the required return now at 11%:

Holding Period	Purchase Price	Market Price	Capital Gain
1 year	₹470.23	₹530.35	₹60.12
2 years	₹470.23	₹528.69	₹58.46
3 years	₹470.23	₹526.84	₹56.62
4 years	₹470.23	₹524.80	₹54.57
5 years	₹470.23	₹522.52	₹52.30
6 years	₹470.23	₹520.00	₹49.77

For example, after one year:

$$Capital\ Gain = 530.35 - 470.23 = ₹60.12$$

compared with only ₹5.83 (476.06 − 470.23) when the required return remains at 14% (Case 11).

D. Final Result

Capital gain after 1 year at 11% required return = ₹60.12 (versus ₹5.83 at 14%)

E. Interpretation

The most important observation is that the capital gain is substantially higher at every holding period when the required return falls from 14% to 11% — most dramatically at short holding periods (₹60.12 versus ₹5.83 after one year), and the gap narrows as the holding period lengthens, since both prices converge to the same ₹520 redemption value at maturity.

This is the fundamental inverse relationship between yields and bond prices at work: YTM ↓ ⟹ Bond Price ↑ ⟹ Capital Gain ↑.

F. Decision / Recommendation

For Suraksha Life Insurance, this shows that a falling-rate environment is unambiguously favourable if the plan is to exit the position early — the capital-gain leg of return rises sharply — even though, as Case 9 showed, the same falling-rate environment simultaneously reduces the interest-on-interest the insurer earns on reinvested coupons. The next case (Case 13) combines both effects into a single total-return figure.

G. Teaching Insight

Teaching Insight: A fall in required return raises bond prices and capital gains, most strongly for near-term exits; this single mechanism — YTM ↓ ⟹ Price ↑ — is the reason bond portfolios post capital gains in rate-cutting cycles, independent of the coupon they carry.

Case 13: Total Return: Coupon, Interest-on-Interest and Capital Gain Combined

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To assemble the three components of bond return — coupon income, interest-on-interest, and capital gain — into a single total-return figure, and to show how the components trade off against each other under different reinvestment/required-return scenarios.

Case Situation

Having computed interest-on-interest (Cases 9-10) and capital gain (Cases 11-12) separately, the investment team at Suraksha Life Insurance now wants to see the complete picture: the total rupee return generated by the bond over each holding period, under both the 14% and 11% rate scenarios, so that the two scenarios can be compared on a like-for-like basis.

Required

1. Assemble the total return (coupon income + interest-on-interest + capital gain) for holding periods of 1 to 6 years, at a reinvestment/required rate of 14%.
2. Repeat the exercise at a reinvestment/required rate of 11%.
3. Explain the trade-off visible between the two scenarios.

Solution

A. Relevant Valuation Principle

A bond's total rupee return over any holding period has three separable components: the coupons themselves, the interest earned by reinvesting those coupons, and any capital gain or loss on sale. Because the same interest-rate variable (the reinvestment/required rate) drives IoI upward but drives capital gain downward when rates are high — and vice versa when rates are low — the two effects partly offset each other, which is precisely why realised return is not simply read off the bond's quoted YTM.

B. Formula

$$\textit{Total Return} = \textit{Coupon Income} + \textit{Interest-on-Interest} + \textit{Capital Gain}$$

C. Substitution and Working

Scenario A — Reinvestment/required rate = 14% (combining Cases 9-11):

Holding Period	Coupon	IoI	Capital Gain	Total Return
1 year	60.00	0.00	5.83	65.83
2 years	120.00	8.40	12.48	140.88
3 years	180.00	26.38	20.06	226.44
4 years	240.00	55.27	28.70	323.97
5 years	300.00	96.61	38.55	435.16

Holding Period	Coupon	IoI	Capital Gain	Total Return
6 years	360.00	152.13	49.77	561.90

For example, for a three-year holding period: Total Return = 180 + 26.38 + 20.06 = ₹226.44.

Scenario B — Reinvestment/required rate = 11% (combining Cases 9-10 and Case 12):

Holding Period	Coupon	IoI	Capital Gain	Total Return
1 year	60.00	0.00	60.12	120.12
2 years	120.00	6.60	58.46	185.06
3 years	180.00	20.53	56.62	257.15
4 years	240.00	42.58	54.57	337.15
5 years	300.00	73.67	52.30	425.97
6 years	360.00	114.77	49.77	524.54

For example, for a three-year holding period: Total Return = 180 + 20.53 + 56.62 = ₹257.15.

D. Final Result

3-year total return at 14%: ₹226.44

3-year total return at 11%: ₹257.15 (higher, despite the lower rate)

E. Interpretation

At the lower rate of 11%, the insurer earns less interest-on-interest, but this is more than offset — at shorter holding periods — by a substantially larger capital gain, since the bond's price benefits from the lower required return. The two effects run in opposite directions: Lower Reinvestment Rate → Lower IoI, but simultaneously Lower Required Return → Higher Bond Price → Higher Capital Gain.

Notice that the gap between the two scenarios narrows sharply as the holding period lengthens (₹54.29 more at 11% for a 1-year hold, but ₹37.36 less at 11% for the full 6-year hold), because the capital-gain advantage of the lower rate is a one-time, front-loaded effect, while the IoI disadvantage compounds over time.

F. Decision / Recommendation

Suraksha Life Insurance's investment team should recognise that for a short holding horizon, a falling-rate scenario is unambiguously beneficial to this bond's total return; but the advantage erodes the longer the bond is held, and by year 6 the 14% scenario in fact delivers a higher total return (₹561.90 versus ₹524.54). The optimal holding period therefore depends on the insurer's view of where rates are headed and how long it intends to hold the position.

G. Teaching Insight

Teaching Insight: Total bond return is always the sum of coupon income, interest-on-interest, and capital gain/loss — and because reinvestment rate and required return move together in practice, a change in market *rates simultaneously helps one component and hurts another. This is the essential reason a bond's realised return can diverge from its quoted YTM.*

Case 14: Realized Yield for a One-Year Holding Period

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute the realized (holding-period) yield on the bond — the actual annualised rate of return earned by the investor — and to show why it can differ sharply from both the bond's YTM and the prevailing reinvestment rate.

Case Situation

Using the total-return figures already computed in Case 13, the investment team at Suraksha Life Insurance wants to convert the one-year total return under each rate scenario into a realized yield — a single percentage figure comparable to the bond's original YTM of 14% — to judge how the insurer actually fared over that year.

Required

1. Calculate the realized yield for a one-year holding period, assuming a 14% reinvestment/required rate.
2. Calculate the realized yield for a one-year holding period, assuming an 11% reinvestment/required rate.
3. Explain why the second figure is so much higher than the reinvestment rate that produced it.

Solution

A. Relevant Valuation Principle

Realized yield is the rate that equates the future value of the amount originally invested to the future value of everything the investor actually received (coupons, interest-on-interest and capital gain/loss) by the end of the holding period. It is the single most accurate measure of what an investor actually earned, as distinct from the yield quoted at purchase.

B. Formula

$$P_0\,(1 + r)^n = \text{Purchase Price} + \text{Total Return}$$

$$\text{i.e., } P_0\,(FVIF_{r,n}) = \text{Future Value of Total Cash Flows}$$

C. Substitution and Working

Example — one-year holding period, reinvestment/required rate 14% (Total Return = ₹65.83 from Case 13):

$$470.23\,(FVIF_{r,1}) = 65.83 + 470.23 = 536.06$$

$$FVIF_{r,1} = 536.06 / 470.23 \approx 1.1400 \implies r = 14\%$$

Example — one-year holding period, reinvestment/required rate 11% (Total Return = ₹120.12 from Case 13):

$$470.23\,(FVIF_{r,1}) = 120.12 + 470.23 = 590.35$$

$$FVIF_{r,1} = 590.35 / 470.23 \approx 1.2554 \implies r \approx 25.54\%$$

D. Final Result

Realized yield (1 year, 14% scenario) = 14%
Realized yield (1 year, 11% scenario) ≈ 25.54%

E. Interpretation

In the 14% scenario, the realized yield exactly equals both the reinvestment rate and the bond's original YTM — unsurprising, since 14% is also the rate at which the bond was originally priced. But in the 11% scenario, although coupons are reinvested at only 11%, the one-year realized yield is a striking 25.54%.

The explanation lies entirely in the capital gain: because the required return fell to 11%, the bond's market price jumped to ₹530.35 within a year, generating a capital gain of ₹60.12 that dwarfs the modest ₹60 coupon and swamps the (as-yet-zero, at a one-year horizon) interest-on-interest. The investor's realized return is therefore driven overwhelmingly by the price effect, not the reinvestment rate that name-checks the scenario.

F. Decision / Recommendation

Suraksha Life Insurance should not mistake the prevailing reinvestment rate for the return the insurer will actually realize on a mark-to-market sale — a falling-rate environment can deliver realized yields far in excess of both the coupon rate and the reinvestment rate, purely through capital appreciation, especially over short holding periods.

G. Teaching Insight

Teaching Insight: Realized yield answers the question 'what did the investor actually earn?', and can diverge sharply from both YTM and the reinvestment rate whenever the bond is sold before maturity at a price that has moved with changing yields. This is the clearest classroom demonstration that YTM is a promise made at purchase, not a guarantee of what will be realized.

Case 15: Realized Yield Across a Full Range of Holding Periods

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To trace how realized yield evolves as the holding period lengthens from 1 to 6 years, and to demonstrate that realized yield converges to YTM only when the reinvestment rate equals YTM, or as the holding period approaches maturity.

Case Situation

To complete the analysis, Suraksha Life Insurance's investment team wants the realized-yield calculation extended across all six possible holding periods (1 through 6 years), for both the 14% and 11% rate scenarios, drawing on the total-return figures already built up in Case 13.

Required

1. Determine the realized yield for holding periods of 1 to 6 years under a 14% reinvestment/required rate.
2. Determine the realized yield for holding periods of 1 to 6 years under an 11% reinvestment/required rate.
3. Explain the pattern observed in the 11% scenario as the holding period lengthens.

Solution

A. Relevant Valuation Principle

Realized yield depends jointly on the coupon, the reinvestment rate, the holding period, and the selling price. When the reinvestment rate equals the bond's YTM, every holding period reproduces the same realized yield (because reinvestment income and capital-gain/loss effects exactly offset the passage of time). When the reinvestment rate differs from YTM, realized yield varies with the holding period, converging toward YTM as the bond approaches maturity and the influence of any one-off capital gain diminishes.

B. Formula

$$P_0\,(1+r)^n = Purchase\ Price + Total\ Return(n)$$

C. Substitution and Working

Applying the realized-yield equation of Case 14 at each holding period, using the total-return figures from Case 13:

Reinvestment Rate	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years
14%	14.0%	14.0%	14.0%	14.0%	14.0%	14.0%
11%	25.54%	18.05%	15.65%	14.47%	13.77%	13.30%

D. Final Result

At 14% reinvestment: realized yield is flat at 14% for every holding period
At 11% reinvestment: realized yield falls from 25.54% (1 yr) to 13.30% (6 yr)

E. Interpretation

When the reinvestment rate exactly equals the bond's YTM (14%), realized yield equals YTM regardless of the holding period — the classic textbook assumption behind quoting YTM as 'the' return on a bond holds exactly in this case.

When the reinvestment rate is below YTM (11%), realized yield starts far above YTM at short holding periods (25.54% at 1 year) because the capital gain from falling rates dominates, then steadily declines toward YTM as the holding period lengthens (18.05% → 15.65% → 14.47% → 13.77%), and falls below YTM by year 6 (13.30%), since the shortfall in interest-on-interest is no longer offset by a growing capital gain (which by maturity has been fully captured at ₹49.77 and cannot grow further).

F. Decision / Recommendation

For Suraksha Life Insurance, this table is a direct guide to exit timing in a falling-rate environment: the earlier the bond is sold after rates fall, the more the investor benefits from the capital-gain windfall; holding to maturity forfeits that windfall entirely and instead exposes the insurer fully to the lower reinvestment rate on coupons, pulling realized yield below the original YTM.

G. Teaching Insight

Teaching Insight: Reinvestment Rate = YTM ⟹ Realized Yield = YTM at any holding period. When the reinvestment rate is lower than YTM, realized yield is elevated by capital gains at short horizons and drifts down toward — and eventually below — YTM as the holding period lengthens toward maturity. Realized yield is therefore a function of coupon, reinvestment rate, holding period and selling price jointly — never of YTM alone.

Part IV — Yield to Call

One case on valuing a callable bond to its call date rather than its final maturity.

Case 16: Yield to Call on a Callable Corporate Bond

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To compute the Yield to Call (YTC) of a callable bond — the return earned if the bond is redeemed at the issuer's option on the call date, rather than held to final maturity.

Case Situation

Himalaya Debt Advantage Fund holds a bond issued by Bharat Renewable Energy Ltd, a wind and solar power generation company. The bond has a face value of ₹1,000 and carries an 11% coupon paid semiannually, with a final maturity of eight years. However, the issuer holds a call option: it may redeem the bond at the end of four years, at a call premium of 8% over face value.

The bond currently trades in the market at ₹1,040. Given the coupon is well above prevailing market rates, the fund's analyst believes the issuer is highly likely to call the bond at the four-year mark rather than let it run to full maturity, and wants to compute the yield the fund would actually earn under that call scenario.

Required

1. Calculate the bond's yield to call (YTC), assuming the bond is called at the end of year 4 at the stated call premium.

Solution

A. Relevant Valuation Principle

Yield to call is computed exactly like yield to maturity, except that the call date replaces the maturity date, and the call price (not the final redemption value) is the terminal cash flow. Whenever a bond is priced above what its yield to maturity alone would suggest is 'fair' relative to prevailing rates — as is common for high-coupon callable bonds — YTC, not YTM, is the more realistic measure of prospective return, since the issuer has a strong economic incentive to call the bond and refinance more cheaply.

B. Formula

$$Market\ Price = C\ (PVIFA\ r, m) + Call\ Price\ (PVIF\ r, m)$$

where C is the semiannual coupon, m is the number of semiannual periods to the call date, and r is the semiannual periodic yield to call.

C. Substitution and Working

Given data:

Item	Value
Face value	₹1,000
Coupon rate (annual)	11%
Semiannual coupon (C)	₹55.00
Call date	End of year 4 (8 semiannual periods)
Call premium	8%
Call price	₹1,080
Current market price	₹1,040

Setting up the valuation equation with 8 semiannual periods to the call date:

$$1{,}040 = 55\,(PVIFA\ r{,}8) + 1{,}080\,(PVIF\ r{,}8)$$

Solving by trial and interpolation (or with a financial calculator) for the semiannual periodic rate r:

$$r \approx 5.68\%\ \text{per half-year}$$

Expressed as a nominal (bond-equivalent) annual rate, doubling the semiannual rate:

$$YTC \approx 5.68\% \times 2 \approx 11.36\%$$

D. Final Result

Semiannual periodic yield ≈ 5.68%
Yield to call (nominal annual) ≈ 11.36%

E. Interpretation

Because the bond pays coupons semiannually, the rate solved directly from the valuation equation is a semiannual periodic rate; it must be doubled (not compounded) to express YTC on the conventional nominal annual, bond-equivalent basis used for quoting Indian and international bond yields. On that basis, YTC works out to approximately 11.36%.

The key distinction from Case 1-2's YTM calculations is the terminal cash flow and horizon used: YTM uses the maturity date and the final redemption value, while YTC uses the call date and the call price — here, eight semiannual periods and ₹1,080, rather than sixteen periods and ₹1,000.

F. Decision / Recommendation

Since Bharat Renewable Energy Ltd's bond carries a coupon well above prevailing market yields, the issuer has a clear economic incentive to call it at the earliest opportunity and refinance at a lower rate. Himalaya Debt Advantage Fund should therefore treat 11.36% — not any yield computed to the eight-year final maturity — as the realistic expected return on this holding, and manage the position (and any duration-matching) on a four-year, not eight-year, horizon.

G. Teaching Insight

Teaching Insight: For a bond trading at a premium and priced well above its likely intrinsic value at the stated coupon, YTC is the economically relevant yield, not YTM — the call date, not the maturity date, is the effective investment horizon, and the call price, not the redemption value, is the correct terminal cash flow to discount.

> ***Editorial / Technical Note:*** *Because the bond's cash flows are semiannual, presenting a single 'YTC' figure requires stating explicitly whether it is a semiannual periodic rate (≈5.68%) or an annualised nominal rate (≈11.36%, obtained by simple doubling, consistent with Indian bond-market convention). This casebook reports the annualised nominal figure as the headline YTC, and shows the semiannual rate explicitly in the working to avoid ambiguity.*

Part V — Duration

Three cases computing Macaulay duration — by direct cash-flow weighting, by the coupon-rate/YTM shortcut formula, and its conversion into modified duration.

Case 17: Macaulay Duration from Discounted Cash Flows

Suraksha Life Insurance Co. Ltd — Fixed Income Portfolio

Learning Objective

To compute a bond's Macaulay duration directly from its schedule of discounted cash flows, and to show how the reinvestment/discount rate used affects the resulting duration figure.

Case Situation

Suraksha Life Insurance's risk team wants to measure the interest-rate exposure of the same 12% coupon, ₹500 face-value bond used throughout Part III (Cases 9-15), by computing its Macaulay duration — the present-value-weighted average time to receipt of its cash flows — at both the 14% and 11% discount rates already used in the earlier analysis.

Required

1. Using the bond's cash flows (₹60 in years 1-5 and ₹580 in year 6, comprising the final coupon plus redemption value), compute Macaulay duration at a 14% discount rate.
2. Repeat the computation at an 11% discount rate, and comment on the difference.

Solution

A. Relevant Valuation Principle

Macaulay duration weights each cash flow's present value by the time at which it is received, then averages these time-weighted present values across the bond's life. It captures, in a single number, the effective average maturity of the bond's cash-flow stream — always shorter than the bond's stated (final) maturity for a coupon-paying bond, since some value is returned earlier via coupons.

B. Formula

$$D = [\,\Sigma_t\, t \cdot PV_t\,] \,/\, [\,\Sigma_t\, PV_t\,]$$

C. Substitution and Working

Cash flows: ₹60 in each of years 1 through 5, and ₹580 (₹60 coupon + ₹520 redemption) in year 6.

Discounting at 14%:

Year	Cash Flow	Present Value	PV × Year
1	60	52.63	52.63
2	60	46.17	92.34

Year	Cash Flow	Present Value	PV × Year
3	60	40.50	121.49
4	60	35.52	142.10
5	60	31.16	155.81
6	580	264.24	1,585.44
Total		**470.23**	**2,149.81**

$$D \text{ (at 14\%)} = 2{,}149.81 / 470.23 \approx 4.57 \text{ years}$$

Discounting the same cash flows at 11% gives a total present value of ₹531.85 and a time-weighted sum of ₹2,479.74:

$$D \text{ (at 11\%)} = 2{,}479.74 / 531.85 \approx 4.66 \text{ years}$$

D. Final Result

Macaulay duration at 14% discount rate ≈ 4.57 years
Macaulay duration at 11% discount rate ≈ 4.66 years

E. Interpretation

Duration (4.57–4.66 years) is meaningfully shorter than the bond's six-year final maturity, because a substantial part of its present value (the ₹60 coupons in years 1-5) is returned well before maturity. The redemption-heavy final cash flow of ₹580 in year 6 still dominates the weighted average, which is why duration remains close to, but below, the full six years.

Duration is slightly longer at the lower 11% discount rate (4.66 years versus 4.57 years) because discounting less heavily gives relatively more weight to the distant year-6 cash flow relative to the near-term coupons.

F. Decision / Recommendation

Suraksha Life Insurance's risk team can use this duration figure — not the bond's six-year stated maturity — as the correct measure of the bond's interest-rate exposure when matching it against policyholder liabilities of a similar duration, a principle developed further in Case 23 (Immunization).

G. Teaching Insight

Teaching Insight: Duration is calculated by multiplying each cash flow's present value by the time at which it is received, summing these products, and dividing by the total present value of all cash flows. It is always less than or equal to a coupon bond's final maturity, and is the correct single-number measure of a bond's effective time horizon for interest-rate risk purposes.

Case 18: Duration Using the Coupon-Rate/YTM Shortcut Formula

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To compute Macaulay duration using the closed-form shortcut formula linking a bond's current yield to its YTM, and to note the special case where the bond is priced at par.

Case Situation

Konkan Bank's treasury desk holds a corporate bond with a face value of ₹2,000, a 7% annual coupon, and five years remaining to maturity, priced to yield 9%. Rather than build out the full cash-flow table each time, the desk's quant analyst wants to apply the algebraic shortcut formula for duration, which uses only the bond's current yield and YTM, to speed up daily duration reporting across the desk's larger bond book.

Required

1. Calculate the bond's price and current yield at the stated YTM of 9%.
2. Using the shortcut duration formula, calculate the bond's Macaulay duration.
3. State how the formula simplifies when a bond is priced at par.

Solution

A. Relevant Valuation Principle

Duration can be computed either the long way — discounting every cash flow individually and time-weighting it — or via a closed-form expression that uses only the ratio of current yield to YTM, the discount rate, and the number of periods to maturity. The two approaches must agree; the shortcut is simply a computational convenience once the bond's price (and hence current yield) is known.

B. Formula

$$P_0 = C\,(PVIFA\ r,N) + F\,(PVIF\ r,N)$$

$$CY = C / P_0$$

$$D = (r\_c/r\_d) \times PVIFA(r\_d,n) \times (1 + r\_d) + [1 - (r\_c/r\_d)] \times n$$

where r_c is the bond's current yield, r_d is the YTM (discount rate), and n is years to maturity. When r_c = r_d (i.e., the bond is priced at par), this simplifies to D = PVIFA(r_d,n) × (1 + r_d).

C. Substitution and Working

Given: Face value = ₹2,000; coupon rate = 7%, so annual coupon C = ₹140; maturity = 5 years; YTM = 9%.

Step 1 — Bond price:

$$P_0 = 140\,(PVIFA\ 9\%,5) + 2{,}000\,(PVIF\ 9\%,5) \approx ₹1{,}844.42$$

Step 2 — Current yield:

$$CY = 140 / 1{,}844.42 \approx 7.59\%$$

Step 3 — Duration, substituting r_c = CY = 7.59% and r_d = YTM = 9%:

$$D = (0.0759 / 0.09) \times PVIFA(9\%,5) \times 1.09 + [1 - (0.0759/0.09)] \times 5$$

$$D \approx 4.409\ years$$

D. Final Result

Bond price ≈ ₹1,844.42
Current yield ≈ 7.59%
Macaulay duration ≈ 4.409 years

E. Interpretation

The duration of 4.409 years is, again, shorter than the bond's five-year maturity, reflecting the value returned early through the 7% coupon. Because this bond trades at a discount (coupon rate 7% below YTM 9%), its current yield (7.59%) sits between the coupon rate and the YTM — a routine feature of discount bonds that the desk's analyst should expect to see rather than treat as an anomaly.

When a bond happens to be priced at par (coupon rate = YTM, so $r_c = r_d$), the formula collapses to the much simpler expression $D = PVIFA(r_d,n) \times (1 + r_d)$ — a useful mental-math shortcut the desk can apply instantly to any bond currently trading at par.

F. Decision / Recommendation

Konkan Bank's treasury desk can use this shortcut formula to compute duration for its entire discount and premium bond book directly from quoted prices and coupons, without re-running a full cash-flow schedule for every security — useful for daily risk reporting across a large portfolio.

G. Teaching Insight

Teaching Insight: The current-yield/YTM ratio formula reproduces approximately the same duration a full cash-flow-weighting calculation would give, but does so directly from price data; its par-value special case, $D = PVIFA(r_d,n)(1+r_d)$, is worth memorising as a quick estimate for any at-par bond.

Case 19: Modified Duration and Estimated Price Sensitivity

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To convert Macaulay duration into modified duration, and to use it to estimate the percentage change in a bond's price for a given change in YTM — the single most widely used measure of interest-rate risk in practice.

Case Situation

The risk desk at Himalaya Debt Advantage Fund holds a 12-year, 9%-coupon government security (face value ₹1,00,000) priced at par (YTM = 9%). Ahead of a scheduled RBI policy announcement, the desk wants a quick, standard estimate of how much the bond's price would move if YTM rose by 1 percentage point — the kind of number quoted routinely on a bond desk's daily risk report.

Required

1. Calculate the bond's Macaulay duration, using the par-bond shortcut from Case 18.
2. Calculate the bond's modified duration.

3. Estimate the percentage change in the bond's price for a 1-percentage-point increase in YTM.

Solution

A. Relevant Valuation Principle

Macaulay duration measures time; modified duration converts it into a direct measure of price sensitivity — the approximate percentage change in bond price for a one-unit change in yield. It is the workhorse risk metric on virtually every fixed-income desk because it translates a somewhat abstract 'weighted average maturity' into an actionable price-risk number.

B. Formula

$$D = PVIFA(r\_d,n) \times (1 + r\_d) \quad [\text{par-bond case}]$$

$$D\_mod = D / (1 + YTM)$$

$$\Delta P/P \approx -D\_mod \times \Delta y$$

C. Substitution and Working

Given: Face value = ₹1,00,000; coupon rate = 9% = YTM, so the bond is priced exactly at par; maturity = 12 years.

Step 1 — Macaulay duration, using the par-bond special case from Case 18 (since coupon rate = YTM here):

$$D = PVIFA(9\%,12) \times 1.09 \approx 7.81 \text{ years}$$

Step 2 — Modified duration (annual compounding):

$$D\_mod = 7.81 / (1 + 0.09) \approx 7.16 \text{ years}$$

Step 3 — Estimated percentage price change for a 1-percentage-point rise in YTM ($\Delta y = 0.01$):

$$\Delta P/P = -7.16 \times 0.01 = -0.0716, \text{ i.e.,} \approx -7.16\%$$

D. Final Result

Modified duration ≈ 7.16 years
Estimated price change for +1% YTM ≈ −7.16%

E. Interpretation

A 1-percentage-point rise in YTM is expected to reduce this bond's price by approximately 7.16%, i.e. by roughly ₹7,160 on the ₹1,00,000 face-value holding; symmetrically (to a first approximation), a 1-point fall in YTM would be expected to raise the price by about the same amount. Modified duration therefore gives the desk an immediate, linear rule of thumb for gauging price risk from any small yield move, without re-running a full valuation.

F. Decision / Recommendation

Himalaya Debt Advantage Fund's risk desk can use this 7.16% figure directly in its daily VaR and scenario reports ahead of the RBI announcement. For larger, non-marginal yield moves, however, the desk should recall Case

4's finding that price changes are not perfectly symmetric — a limitation addressed formally with convexity in Cases 24-25.

G. Teaching Insight

Teaching Insight: $D_{mod} = D / (1+YTM)$ converts Macaulay duration into a percentage-price-sensitivity measure, $\Delta P/P \approx -D_{mod} \times \Delta y$. It is a first-order (linear) approximation that is highly accurate for small yield changes but progressively less accurate for larger ones — the exact gap that convexity is designed to close.

> ***Editorial / Technical Note:*** *This modified-duration illustration (a par bond with $D \approx 7.81$ years at a 9% YTM, giving $D_{mod} \approx 7.16$ and a −7.16% price-sensitivity estimate) consolidates what appeared as two separate, numerically identical illustrations in the original source material — once introducing modified duration directly after Macaulay duration, and again as the opening illustration of the advanced interest-rate-risk module. They have been merged into this single case to avoid redundancy in the casebook, while both original teaching contexts (a bridge from duration, and an anchor for the elasticity/effective-duration/convexity sequence that follows) are preserved by this case's placement at the end of Part V.*

Part VI — Interest-Rate Risk

Three cases measuring bond price sensitivity to yield changes — interest rate elasticity, its link to duration, and effective duration computed from directly repriced bonds.

Case 20: Interest Rate Elasticity of a Par Bond

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To measure a bond's interest rate elasticity — the percentage change in price for a given percentage change in YTM — as a scale-free alternative to quoting price sensitivity in absolute terms.

Case Situation

Continuing its RBI-announcement risk assessment, Himalaya Debt Advantage Fund's risk desk wants to express the same 12-year, 9%-coupon, par-priced government security's interest-rate risk not just as a rupee or percentage price move, but as an elasticity figure — directly comparable, in principle, to the price elasticities used elsewhere in finance and economics.

Required

1. Calculate the bond's interest rate elasticity if YTM rises from 9% to 10%.

Solution

A. Relevant Valuation Principle

Interest rate elasticity (also called bond volatility) expresses price sensitivity as a ratio of two percentage changes — percentage change in price divided by percentage change in YTM — rather than price change divided by the absolute (percentage-point) change in yield used in the modified-duration formula. It is a direct measure of a bond's price sensitivity, closely related to, but scaled differently from, modified duration.

B. Formula

$$IE = (\%\ \text{Change in Bond Price}) / (\%\ \text{Change in YTM}) = (\Delta P/P_0) / (\Delta YTM/YTM)$$

C. Substitution and Working

Given: Face value = ₹1,00,000; coupon rate = 9% = YTM (bond priced at par); maturity = 12 years.

Step 1 — Initial price (at par):

$$P_0 = ₹1{,}00{,}000$$

Step 2 — Price at YTM = 10%:

$$P_1 = ₹93{,}186.31 \quad \Longrightarrow \quad \Delta P = 93{,}186.31 - 1{,}00{,}000 = -₹6{,}813.69$$

Step 3 — Percentage change in price:

$$\Delta P/P_0 = -6{,}813.69 \,/\, 1{,}00{,}000 = -6.81\%$$

Step 4 — Percentage change in YTM:

$$\Delta YTM/YTM = (0.10 - 0.09) \,/\, 0.09 = 11.11\%$$

Step 5 — Interest rate elasticity:

$$IE = -6.81\% \,/\, 11.11\% = -0.613$$

D. Final Result

Interest rate elasticity ≈ −0.613

E. Interpretation

The negative sign confirms the fundamental inverse relationship between bond price and YTM: YTM ↑ ⟹ Bond Price ↓. The magnitude, 0.613, indicates that for every 1% relative change in YTM, this bond's price moves by roughly 0.613% in the opposite direction — a scale-free measure that can be compared across bonds of very different starting prices and yields.

F. Decision / Recommendation

Himalaya Debt Advantage Fund can use interest rate elasticity to compare risk across bonds trading at very different yield levels — a use case where modified duration's rupee/percentage-point framing is less directly comparable across issues with dissimilar YTMs.

G. Teaching Insight

Teaching Insight: Interest rate elasticity re-expresses price sensitivity as a ratio of percentage changes rather than a percentage change per percentage-point of yield, and — as Case 21 shows next — it is directly and simply related to duration, the two measures being two views of the same underlying price-yield relationship.

Case 21: Linking Duration to Interest Rate Elasticity

Himalaya Debt Advantage Fund — Risk Management Desk

Learning Objective

To derive interest rate elasticity directly from a bond's duration, demonstrating the algebraic link between the two measures of interest-rate risk.

Case Situation

Having computed interest rate elasticity directly from price changes in Case 20, the risk desk now wants to verify the figure using an entirely separate route — computing it from the bond's duration — both as a cross-check and to build the desk's intuition for how duration and elasticity relate.

Required

1. Calculate the duration of the same 12-year, 9%-coupon government security (priced at par).

2. Use the duration-elasticity relationship to compute interest rate elasticity, and compare it with the directly computed figure from Case 20.

Solution

A. Relevant Valuation Principle

Because both duration and elasticity are derived from the same price-yield relationship, they are linked by a simple algebraic identity involving the level of YTM. This identity lets an analyst move freely between the two measures without recomputing bond prices from scratch.

B. Formula

$$D = PVIFA(r\_d,n) \times (1 + r\_d) \quad [par\text{-}bond\ case]$$

$$IE = D \times YTM \,/\, (1 + YTM)$$

C. Substitution and Working

Step 1 — Duration, using the par-bond shortcut (coupon rate = YTM = 9%, n = 12 years):

$$D = PVIFA(9\%,12) \times 1.09 \approx 7.81$$

Step 2 — Interest rate elasticity from duration:

$$IE = 7.81 \times (0.09 \,/\, 1.09) \approx 0.645$$

D. Final Result

> **Duration ≈ 7.81 years**
> **Duration-based elasticity ≈ 0.645 (magnitude)**

E. Interpretation

The duration-based elasticity of 0.645 is close to, though not identical to, the 0.613 computed directly from price changes in Case 20; the small gap arises because the direct calculation captures the bond's true (convex) price-yield curve over a full 1-percentage-point move, while the duration-based formula is a linear approximation valid strictly at a single point on that curve.

This relationship, IE = D × YTM/(1+YTM), shows that at a given YTM, elasticity and duration always move together — a bond with higher duration always has higher (more negative) interest rate elasticity, and vice versa.

F. Decision / Recommendation

The risk desk can treat duration as the primary risk metric for day-to-day monitoring (since it is additive across a portfolio and easy to interpret in time units), while using this identity to translate into elasticity terms whenever a scale-free comparison across differently-yielding bonds is required.

G. Teaching Insight

Teaching Insight: Duration-based elasticity is a convenient way of relating duration to bond price sensitivity, but because the true bond price-yield relationship is convex, duration-based estimates are approximations that lose accuracy for larger, non-marginal changes in YTM — exactly the limitation that convexity (Cases 24-25) is built to address.

Case 22: Effective Duration from an Observed Yield Shock

Konkan Bank Ltd — Treasury and Investments Desk

Learning Objective

To compute effective duration directly from observed bond prices under a small up- and down-shock to yield, a method that remains valid even when a bond's cash flows themselves may change with yield.

Case Situation

Konkan Bank's treasury desk holds an eight-year, 7.5%-coupon bond (face value ₹1,000), currently priced at par since its YTM also stands at 7.5%. As part of a routine risk-model validation, the desk's quant analyst reprices the bond under a small parallel yield shock of 25 basis points in each direction — down to 7.25% and up to 7.75% — and wants to derive the bond's effective duration directly from the resulting prices, rather than from a formula based on the bond's stated cash flows.

Required

1. Given that the bond reprices to ₹1,014.78 when the yield falls to 7.25%, and to ₹985.50 when the yield rises to 7.75%, calculate the bond's effective duration.

Solution

A. Relevant Valuation Principle

Effective duration measures price sensitivity by directly repricing the bond (using its full valuation model) under small up- and down-shifts in yield, rather than relying on a closed-form duration formula. This makes it the appropriate measure for instruments whose future cash flows themselves may change as yields move — for example, callable or putable bonds, or mortgage-backed securities — where Macaulay/modified duration formulas (which assume fixed cash flows) can be misleading.

B. Formula

$$D\_eff = (P\_down - P\_up) / (2 \times P_0 \times \Delta y)$$

C. Substitution and Working

Given data:

Item	Value
Initial price, P_0 (YTM = 7.5%)	₹1,000.00
Price when yield falls to 7.25%, P_down	₹1,014.78

Item	Value
Price when yield rises to 7.75%, P_up	₹985.50
Yield shock, Δy	25 basis points = 0.25% = 0.0025

$$D\_eff = (1{,}014.78 - 985.50) / (2 \times 1{,}000 \times 0.0025)$$

$$D\_eff = 29.28 / 5.00 \approx 5.86 \text{ years}$$

D. Final Result

Effective duration ≈ 5.86 years

E. Interpretation

The bond's price rises by more (₹14.78) when yield falls by 25bp than it falls (₹14.50) when yield rises by the same 25bp — the same convexity-driven asymmetry seen in Case 4, here at a much smaller yield increment. Effective duration of 5.86 years indicates that the bond's price will move by roughly 5.86% for a 100-basis-point parallel shift in yield, in either direction, over this small range.

F. Decision / Recommendation

Konkan Bank's treasury desk should use effective duration, computed this way from actual repriced values, for any instrument in its book whose cash flows could change with yield (callable bonds, embedded-option structures); for plain-vanilla, fixed-cash-flow bonds like this one, effective duration and modified duration will normally be very close, as a useful cross-check on the desk's valuation models.

G. Teaching Insight

Teaching Insight: Effective duration is computed by repricing a bond under small symmetric yield shocks and reading off the resulting price sensitivity — it makes no assumption that the bond's cash flows are fixed, which is precisely why it is the standard duration measure for callable bonds, putable bonds and mortgage-backed securities.

Part VII — Immunization

One case applying duration matching to immunize a stream of future liabilities using a two-bond zero-coupon portfolio.

Case 23: Immunizing a Five-Year Gratuity Payout

Approved Gratuity Trust of an Indian Manufacturing Company

Learning Objective

To compute the duration of a stream of fixed future liabilities, and to determine how a portfolio of zero-coupon bonds should be allocated so that the assets' duration matches the liabilities' duration — the core technique of duration-based immunization.

Case Situation

The trustees of the approved gratuity trust of an Indian manufacturing company must fund a retiring employee's gratuity, to be paid out in five equal annual instalments of ₹50,000 each, starting one year from now. The trustees want to invest the corpus today in a way that is protected (immunized) against future interest-rate movements, rather than simply matching the final payment date.

Two zero-coupon government bonds are available for the trust to invest in: one maturing in 2 years, and one maturing in 6 years. The applicable required rate of return on the corpus is 10%.

Required

1. Calculate the duration of the liability stream (the five annual ₹50,000 payments).
2. Determine what percentage of the corpus should be invested in the 2-year zero-coupon bond and what percentage in the 6-year zero-coupon bond so that the portfolio's duration matches the liability duration.

Solution

A. Relevant Valuation Principle

Duration-based immunization matches the duration — not the maturity — of the asset portfolio to the duration of the liabilities. Because a portfolio's duration is a value-weighted average of the durations of its holdings, and because a zero-coupon bond's duration equals its maturity exactly, a two-bond portfolio can be tuned to any duration between the two bonds' maturities simply by adjusting the allocation between them. When asset duration equals liability duration, the portfolio's value and the liabilities' present value respond to a (small, parallel) shift in rates in the same way, protecting the trust's ability to meet its obligations regardless of which way rates move.

B. Formula

$$D\_L = [\, \Sigma_t \; t \cdot PV_t \,] \,/\, [\, \Sigma_t \; PV_t \,]$$

$$D\_Portfolio = X \times D(\text{2-year bond}) + (1 - X) \times D(\text{6-year bond})$$

C. Substitution and Working

Step 1 — Present value of the liabilities, discounted at 10%:

Year	Cash Flow	PV at 10%	PV × Year
1	₹50,000	₹45,454.55	₹45,454.55
2	₹50,000	₹41,322.31	₹82,644.63
3	₹50,000	₹37,565.74	₹1,12,697.22
4	₹50,000	₹34,150.67	₹1,36,602.69
5	₹50,000	₹31,046.07	₹1,55,230.33
Total	**₹2,50,000**	**₹1,89,539.34**	**₹5,32,629.42**

$$D\_L = 5{,}32{,}629.42 \,/\, 1{,}89{,}539.34 \approx 2.81 \text{ years}$$

Step 2 — Allocation between the 2-year and 6-year zero-coupon bonds. Let X = proportion invested in the 2-year bond, so (1 − X) is invested in the 6-year bond. Since a zero-coupon bond's duration equals its own maturity, the portfolio-duration condition is:

$$2X + 6(1 - X) = 2.81$$

$$2X + 6 - 6X = 2.81 \implies 6 - 4X = 2.81 \implies 4X = 3.19 \implies X \approx 0.7975$$

D. Final Result

Duration of the liability stream ≈ 2.81 years

Allocate ≈ 79.75% of the corpus to the 2-year zero-coupon bond

Allocate ≈ 20.25% of the corpus to the 6-year zero-coupon bond

E. Interpretation

The liability stream, though paid out over five years, has a duration of only about 2.81 years, since the earlier instalments (discounted less) carry more weight in the average than the later ones. To replicate this duration with only two available zero-coupon instruments (2-year and 6-year), the trust must weight its holdings more heavily toward the shorter bond — about 79.75% in the 2-year bond and 20.25% in the 6-year bond.

F. Decision / Recommendation

The trustees should invest the corpus (₹1,89,539.34, the present value of the liabilities) in this roughly 80:20 split between the 2-year and 6-year zero-coupon bonds. This immunizes the trust's ability to meet the five gratuity instalments against small, parallel shifts in interest rates: a rate rise would reduce the present value of the remaining liabilities but also reduce the value of the bond portfolio by a matching amount (and vice versa for a rate fall), leaving the trust's funded status broadly protected either way. The trustees should note that this protection holds only for small, parallel shifts, and that the allocation will need to be rebalanced periodically as time passes and durations drift (duration-based immunization is a dynamic, not a set-and-forget, strategy).

G. Teaching Insight

Teaching Insight: Immunization matches Duration(Assets) = Duration(Liabilities), not Maturity(Assets) = Maturity(Liabilities). Because zero-coupon bond duration equals maturity exactly, a two-bond zero-coupon portfolio can be tuned to any target duration between the two maturities by adjusting the allocation weights — the standard technique for funding a known, fixed liability schedule.

Part VIII — Convexity

Two cases computing bond convexity and using the duration-plus-convexity approximation to improve on duration-only price-sensitivity estimates for large yield moves.

Case 24: Computing Bond Convexity for a High-Coupon Bond

Himalaya Debt Advantage Fund — Fixed Income Desk

Learning Objective

To compute a bond's convexity — the second-order measure of price sensitivity that captures the curvature of the price-yield relationship left unexplained by duration alone.

Case Situation

Himalaya Debt Advantage Fund holds a five-year, 12%-coupon corporate bond (face value ₹2,000), currently priced to yield 9%. Having already used modified duration to estimate this bond's price sensitivity, the fund's quant analyst now wants to compute its convexity, so that a more accurate, curvature-adjusted price-sensitivity estimate can be produced for larger yield moves (a technique applied in Case 25).

Required

1. Calculate the convexity of the bond.

Solution

A. Relevant Valuation Principle

Convexity measures the rate at which a bond's duration itself changes as yield changes — equivalently, the curvature of the price-yield relationship. It is computed by weighting each cash flow's present value not just by time (as in duration) but by a function of time-squared, capturing how much more (or less) sensitive distant cash flows are to yield changes than duration's linear approximation alone would suggest.

B. Formula

$$Convexity = [\,\Sigma_t\, PV_t\,(t^2 + t)\,] \,/\, [\,P_0\,(1 + y)^2\,]$$

C. Substitution and Working

Given: Face value = ₹2,000; coupon rate = 12%, so annual coupon C = ₹240; maturity = 5 years; YTM = 9%.

Step 1 — Cash flows:

Year	Cash Flow
1	₹240
2	₹240
3	₹240

Year	Cash Flow
4	₹240
5	₹2,240

Step 2 — Present values at 9% YTM:

Year	Cash Flow	PV
1	240	220.18
2	240	202.00
3	240	185.32
4	240	170.02
5	2,240	1,455.85
Total		**2,233.38**

Step 3 — Convexity numerator: for each cash flow, compute $PV_t \times (t^2 + t)$:

Year (t)	PV	$t^2 + t$	$PV \times (t^2 + t)$
1	220.18	2	440.37
2	202.00	6	1,212.02
3	185.32	12	2,223.89
4	170.02	20	3,400.44
5	1,455.85	30	43,675.39
Total			**50,952.10**

Step 4 — Convexity:

$$Convexity = 50{,}952.10 / [2{,}233.38 \times (1.09)^2] \approx 19.20$$

D. Final Result

Bond price (9% YTM) ≈ ₹2,233.38
Convexity ≈ 19.20

E. Interpretation

A convexity of 19.20 means that, beyond the linear price sensitivity already captured by duration, this bond's price-yield curve bends further in the investor's favour as yields move — the price falls a little less than duration alone would predict when yields rise, and rises a little more than duration predicts when yields fall (the same qualitative asymmetry demonstrated back in Case 4, now given a precise numerical measure).

F. Decision / Recommendation

Himalaya Debt Advantage Fund's analyst should carry this convexity figure forward into any price-sensitivity estimate that spans a large yield move (of the kind illustrated for a different bond in Case 25); for small moves, duration alone remains an adequate approximation and the convexity term contributes negligibly.

G. Teaching Insight

Teaching Insight: Convexity, like duration, is computed by present-value-weighting each cash flow — but weighted by ($t^2 + t$) rather than t — capturing the curvature of the price-yield relationship. Two bonds with identical duration can have different convexity; the more convex bond is, all else equal, the more attractive holding, since it gains more on a rally than it loses on a sell-off of equal size.

Case 25: Duration-Convexity Approximation for a Large Yield Move

Suraksha Life Insurance Co. Ltd — Long-Duration G-Sec Portfolio

Learning Objective

To compare a duration-only price estimate against a duration-plus-convexity estimate for a large change in yield, and to show how much more accurate the convexity-adjusted estimate is relative to the bond's true, fully revalued price.

Case Situation

Suraksha Life Insurance holds a long-dated, 25-year, 7.5%-coupon government security in its policyholders' fund, purchased at par (face value ₹1,00,000) when its YTM also stood at 7.5%. Following a sharp tightening in the interest-rate environment, the bond's YTM has since risen to 9.5% — a large, 2-percentage-point move that the insurer's ALM desk wants to analyse carefully, since large yield swings are exactly where a simple duration-only estimate is known to be least reliable.

The bond's modified duration at its original 7.5% yield is 11.15 years, and its convexity is 195.56. The desk wants to compare what duration alone would have predicted, what duration-plus-convexity predicts, and how both compare with the bond's actual, fully revalued price at the new 9.5% yield (₹81,124.84).

Required

1. Estimate the percentage decline in the bond's price using modified duration alone.
2. Estimate the percentage decline using the duration-plus-convexity approximation.
3. Compare both estimates against the bond's actual percentage price decline.

Solution

A. Relevant Valuation Principle

The duration-only approximation is a straight-line (tangent-line) estimate of price sensitivity, and because the true price-yield relationship is convex, this straight-line estimate systematically overstates the price decline for a rise in yield (and understates the price gain for a fall in yield). Adding a convexity term corrects for this curvature and produces a materially more accurate estimate, especially for large yield moves such as this 2-percentage-point shift.

B. Formula

$$\text{Duration-only: } \Delta P/P \approx -D\_mod \times \Delta y$$

$$\text{Duration + Convexity: } \Delta P/P \approx -D\_mod \times \Delta y + ½ \times \text{Convexity} \times (\Delta y)^2$$

C. Substitution and Working

Given: 25-year, 7.5%-coupon government security, initially priced at par (₹1,00,000) at YTM = 7.5%; modified duration = 11.15 years; convexity = 195.56; YTM rises from 7.5% to 9.5%, so Δy = 0.02.

Step 1 — Duration-only estimate:

$$\Delta P/P = -11.15 \times 0.02 = -0.2229, \text{ i.e., } \approx -22.29\%$$

Step 2 — Duration-plus-convexity estimate:

$$\Delta P/P = -11.15 \times 0.02 + ½ \times 195.56 \times (0.02)^2$$

$$= -0.2229 + 0.0391 = -0.1838, \text{ i.e., } \approx -18.38\%$$

Step 3 — Actual price change. The bond's fully revalued price at YTM = 9.5% is ₹81,124.84, against an initial price of ₹1,00,000:

$$\text{Actual \% Change} = (81{,}124.84 - 1{,}00{,}000) / 1{,}00{,}000 \times 100 \approx -18.88\%$$

Method	Estimated Price Decline
Duration only	−22.29%
Duration + convexity	−18.38%
Actual (fully revalued)	−18.88%

D. Final Result

Duration-only estimate: −22.29%
Duration + convexity estimate: −18.38%
Actual price decline: −18.88%

E. Interpretation

The duration-only estimate overstates the actual price decline by a wide margin — 22.29% predicted versus 18.88% actual, an error of 3.41 percentage points — because it assumes a linear price-yield relationship over what is, in reality, a 2-percentage-point move on a highly convex, long-duration bond. Adding the convexity term brings the estimate to 18.38%, within 0.50 percentage points of the true decline — a substantial improvement in accuracy for a large, non-marginal yield shock.

F. Decision / Recommendation

For Suraksha Life Insurance's ALM desk, this confirms that long-duration government securities in the policyholders' fund should be marked and stress-tested using duration-plus-convexity, not duration alone, whenever the scenario involves a large parallel yield shift (such as a full RBI policy cycle) — the duration-only method would otherwise lead the desk to overstate potential mark-to-market losses on its long-bond book by a material margin, distorting capital and hedging decisions.

G. Teaching Insight

Teaching Insight: Duration + Convexity is a materially better approximation than Duration alone for large yield changes, precisely because it captures the curvature of the true price-yield relationship. The larger the yield move and the longer a bond's duration, the more the convexity term matters — which is exactly why convexity receives the most attention for long-duration government-security and pension-fund portfolios.

Consolidated Learning Map

The twenty-five cases in this book move through a single coherent progression, from the basic mechanics of bond pricing to the advanced toolkit portfolio and risk managers use to measure and manage interest-rate exposure:

Bond Price → YTM → Term Structure → Bond Return → Yield to Call → Duration → Interest-Rate Risk → Immunization → Convexity

Parts I–III (Cases 1–15) establish how a bond is priced, why its price moves the way it does, and what an investor actually earns over a chosen holding period — culminating in the crucial distinction between a bond's quoted YTM and its realized yield.

Part IV (Case 16) extends bond valuation to callable structures, where the call date and call price, not the final maturity, are the economically relevant terms.

Parts V–VI (Cases 17–22) build the duration-based toolkit for measuring interest-rate risk, moving from Macaulay duration through modified duration, interest rate elasticity, and effective duration.

Part VII (Case 23) applies this toolkit to a real portfolio-management problem — immunizing a stream of future liabilities against interest-rate movements.

Part VIII (Cases 24–25) completes the sequence with convexity, the refinement that makes duration-based price-sensitivity estimates materially more accurate for large yield moves.